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Mini Project Final Review

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Mini Project Work

Topic: FILE ENCRYPTOR AND DECRYPTOR

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Abstract

File Encryption and Decryption

The growing reliance on digital technology has heightened the importance of securing sensitive information. This project provides an effective solution to protect data through encryption and decryption. File encryption and decryption are essential security measures that protect sensitive data from unauthorized access. Encryption involves converting plaintext files into unreadable ciphertext using an encryption algorithm and key. Decryption reverses the process, restoring the original plaintext file. This ensures confidentiality, integrity, and authenticity of data.

Introduction

1. **Importance of Data Security** : In today's digital era, sensitive data is often stored and shared electronically, making it vulnerable to breaches and unauthorized access. File encryption and decryption provide a robust mechanism to ensure confidentiality, integrity, and authenticity of data.
2. **What is Encryption and Decryption?**:Encryption: Converts plaintext into unreadable ciphertext using algorithms and keys. Decryption Restores ciphertext back to plaintext, requiring the correct key for access.
3. **Relevance**: Applicable in personal, academic, and professional settings. Essential for protecting personal files, and organizational data from cyber threats.

Existing Apps

1. **WinRAR/7-Zip**:File compression tools with basic encryption capabilities. Limitation: Weak encryption methods; not designed for modern cryptographic standards.
2. **VeraCrypt** : Provides disk encryption and supports multiple encryption algorithms.
3. **Limitation**: Complex user interface; requires significant user expertise and Restricted to specific operating systems; limited file-based encryption.

Limitations of Existing Apps

1. **Complexity:** Many apps require technical knowledge, making them less user-friendly for non-experts.
2. **Lack of Cross-Platform Support**: Limited compatibility across operating systems (Windows, macOS, Linux).
3. **Weak Key Management:** Poor handling of encryption keys can compromise security.

Necessity of the Proposed Project

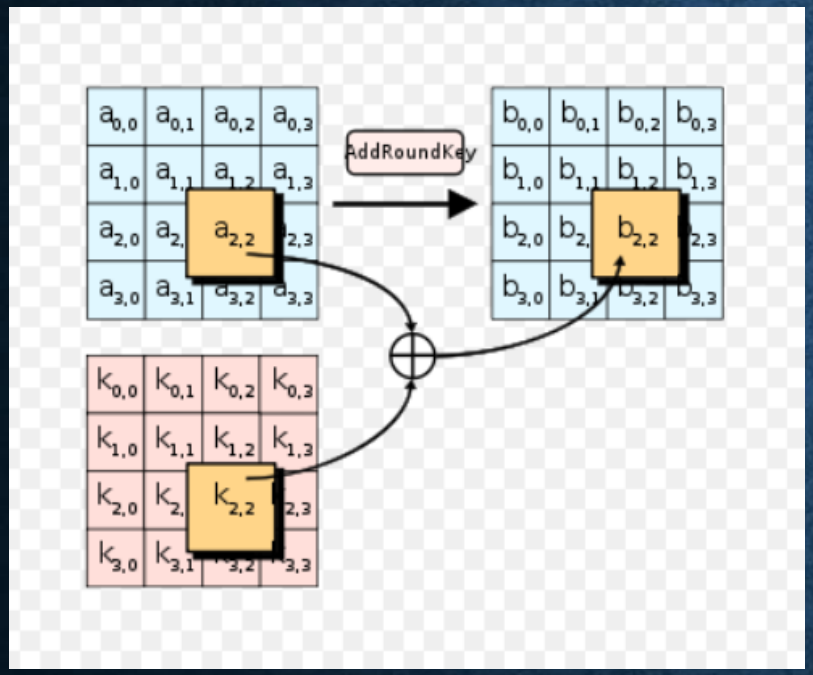
1. **Ease of Use** : A user-friendly interface ensures accessibility for all, including non-technical users.
2. **Improved Security** : Utilizes advanced cryptographic algorithms to safeguard sensitive data.
3. **Unique Key Generation** : Generates distinct keys for each file, enhancing security.
4. **Cross-Platform Compatibility** : Designed to work seamlessly across multiple operating systems
5. **Affordability**: Provides essential encryption features without costly subscriptions.

Literature Review

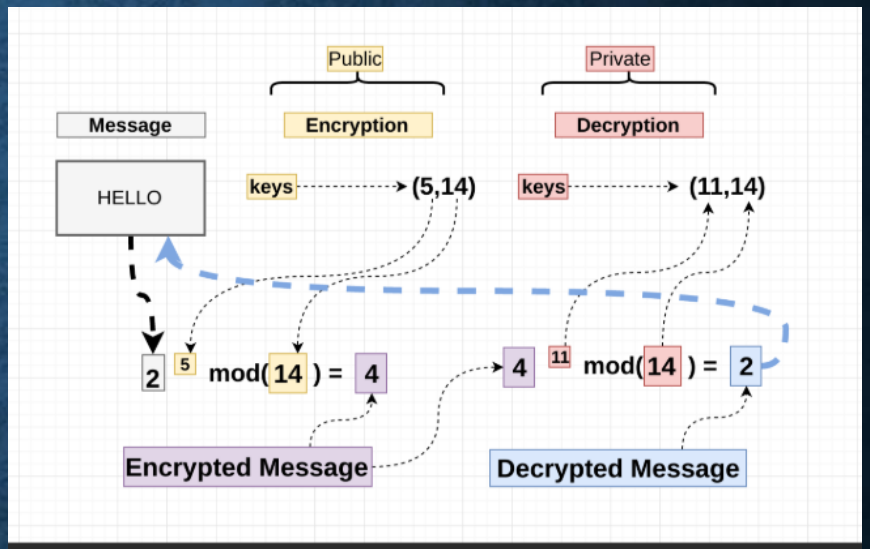
1. **Available Tools** : VeraCrypt: Strong disk encryption, complex usage

2. **AxCrypt** : File encryption with limited free features

2. **Technology Choice** : Using modern cryptographic libraries (e.g., AES, RSA).



AES stands for Advanced Encryption Standard.



RSA stands for Rivest-Shamir-Adleman. It's a public-key encryption algorithm widely used for secure data transmission.

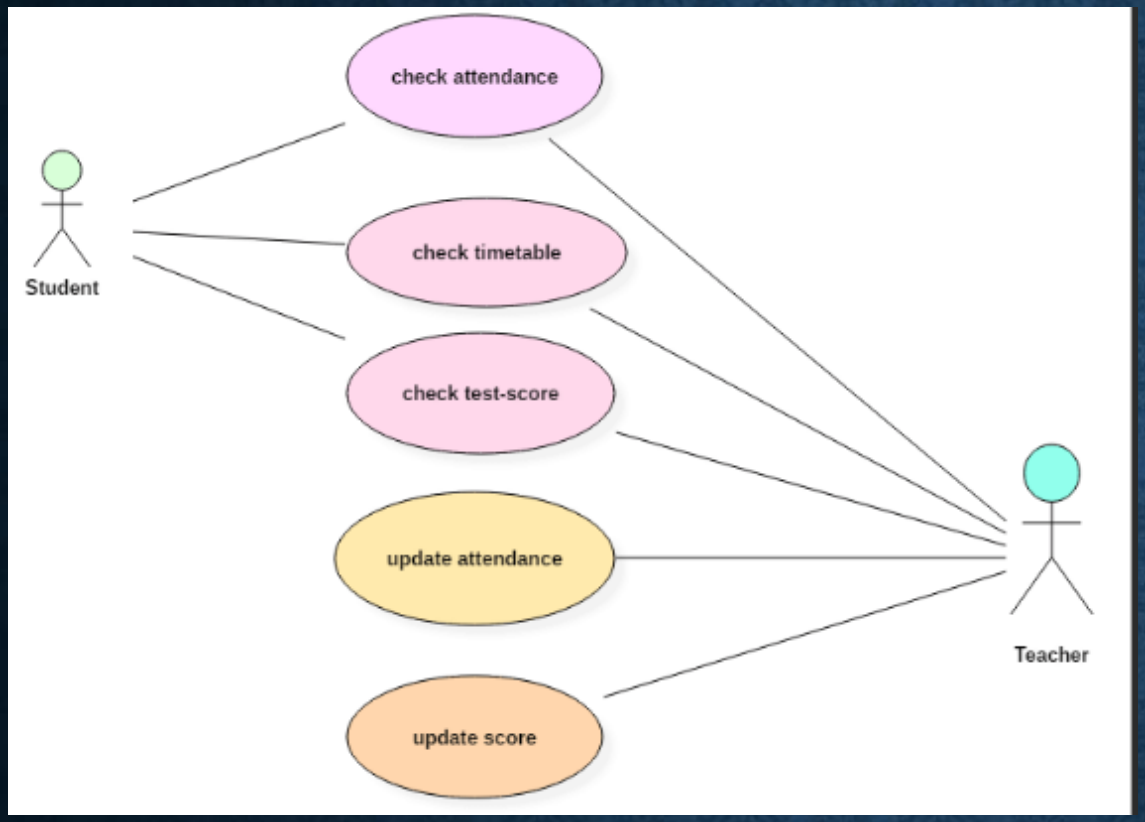
Analysis and Design

1. **File Encryption** : Allow users to select files for encryption . Use advanced cryptographic algorithms (e.g., AES-256) for secure encryption.
2. **File Decryption** : Provide decryption functionality that restores the original file using the correct key.
3. **Key Generation and Management** : Automatically generate unique encryption keys for each file . Ensure secure storage of keys (if saved locally) or provide download options.
4. **User Interface** : Provide a user-friendly interface for file selection, encryption, and decryption . Display success/failure notifications for each operation.
5. **Error Handling** : Detect and notify users of invalid or corrupted files during decryption.
6. **Platform Independence** : Support multiple platforms (Windows, Linux, macOS).

Non-Functional Requirements

1. **Security** : Use state-of-the-art encryption algorithms and secure key storage methods . Protect against brute force and replay attacks.
2. **Performance** : Ensure encryption and decryption processes are efficient and do not cause significant delays.
3. **Usability** : Provide an intuitive interface with minimal learning curve . Offer clear error messages and guidance for troubleshooting.
4. **Scalability**: Handle various file sizes efficiently, from small text files to large media files.
5. **Reliability** : Ensure consistent performance without data corruption or loss . Prevent crashes or errors during critical processes.
6. **Portability**: Ensure compatibility with different operating systems and environments.
7. **Maintainability** : Modular codebase for easy updates and bug fixes. Clear documentation for future enhancements.

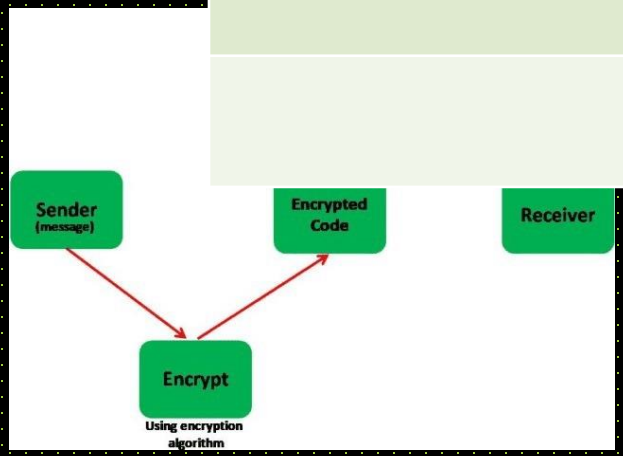
Use Case Diagram



1. Actor : User: Interacts with the system to encrypt or decrypt files.

2. Use Cases : Select File: User selects the file they want to encrypt or decrypt . Encrypt File: Encrypts the selected file using a generated key.

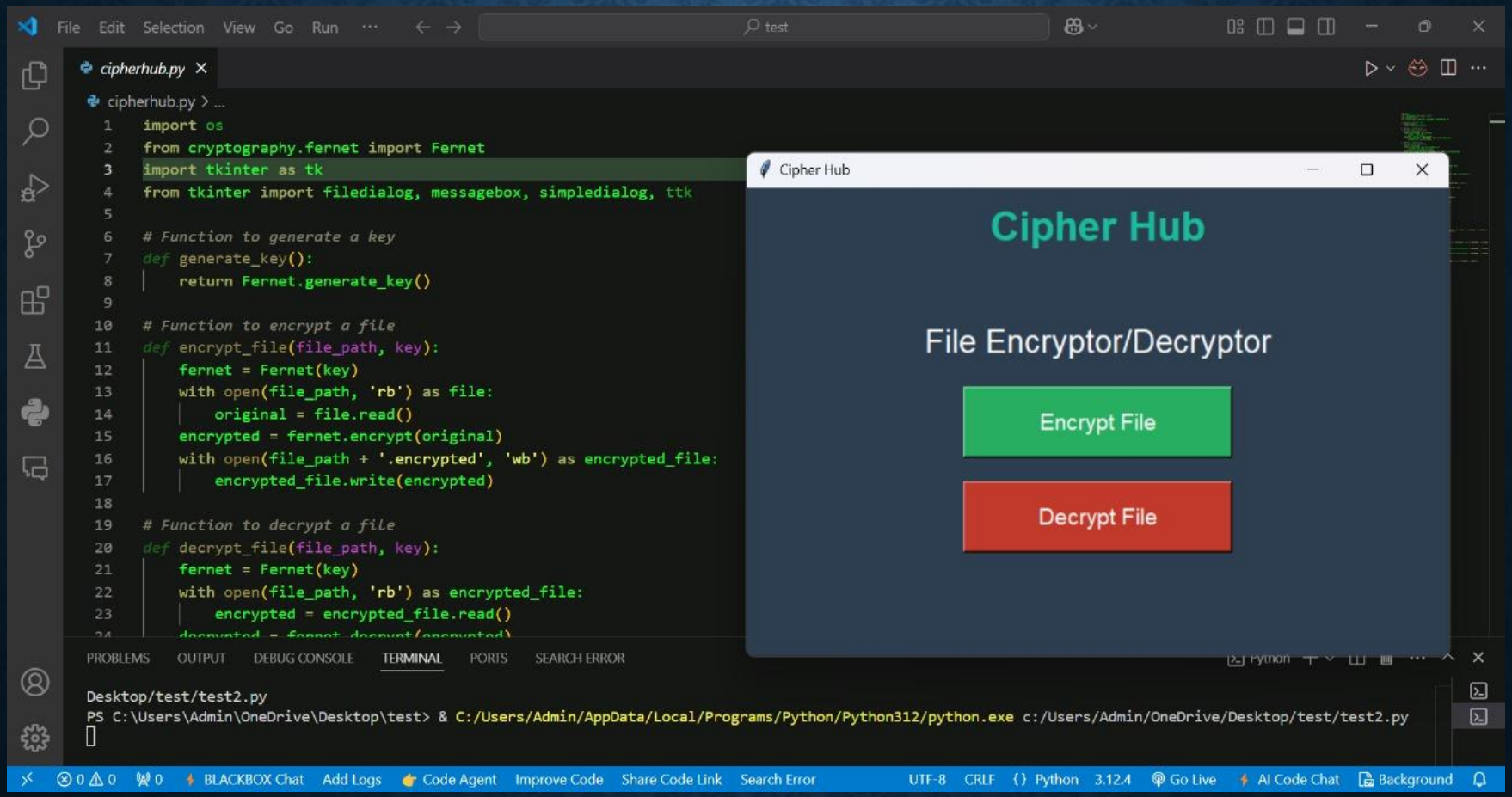
3. Decrypt File: Decrypts the file using the corresponding key.

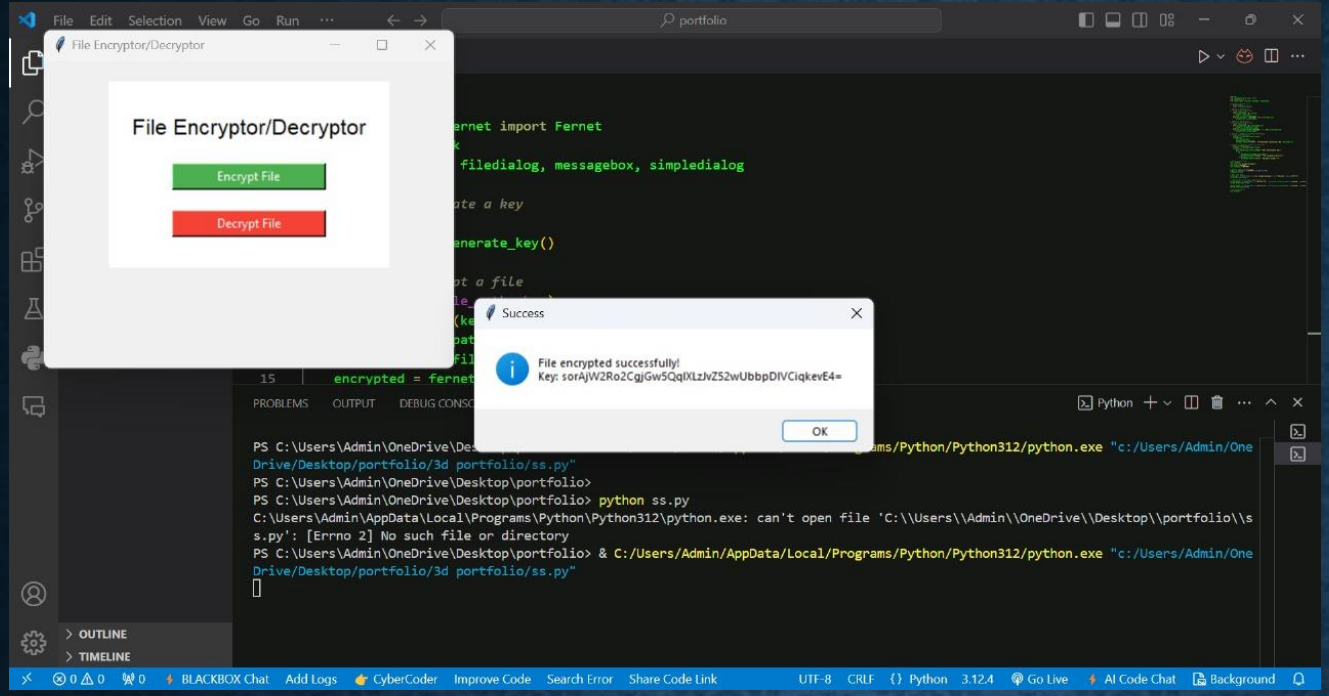


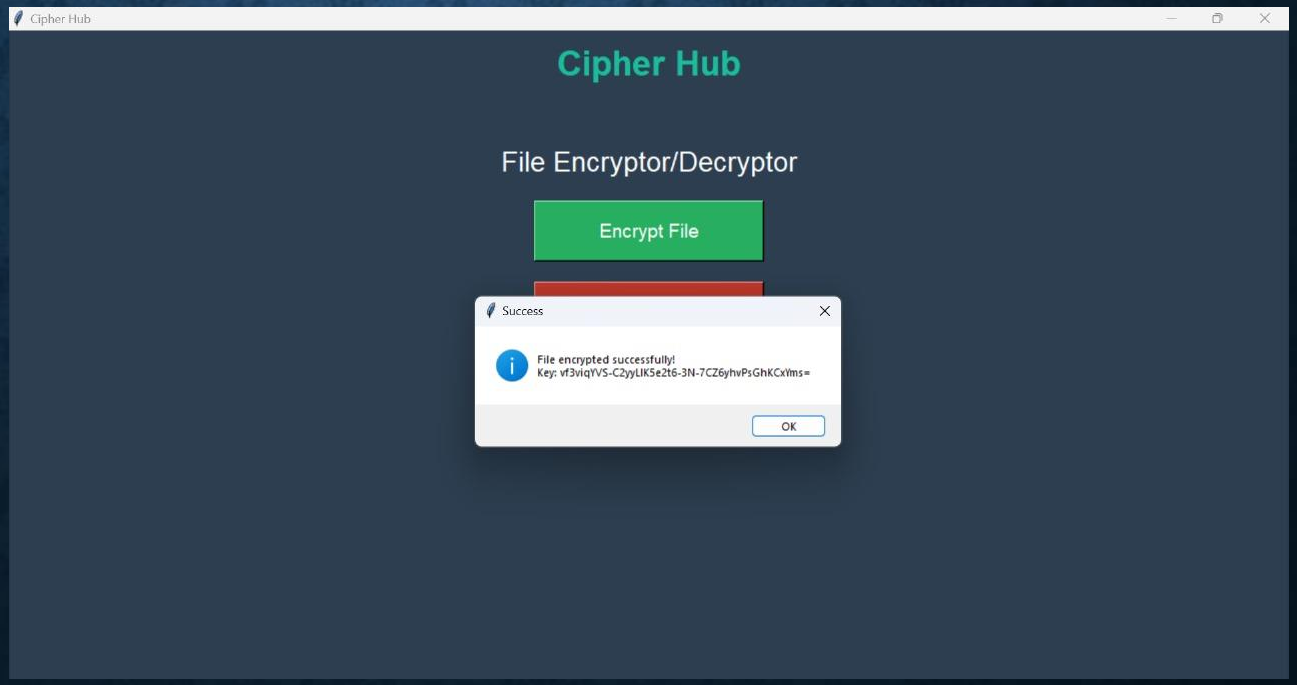
ates
actions like selecting files, generating keys, encrypting, and decrypting .

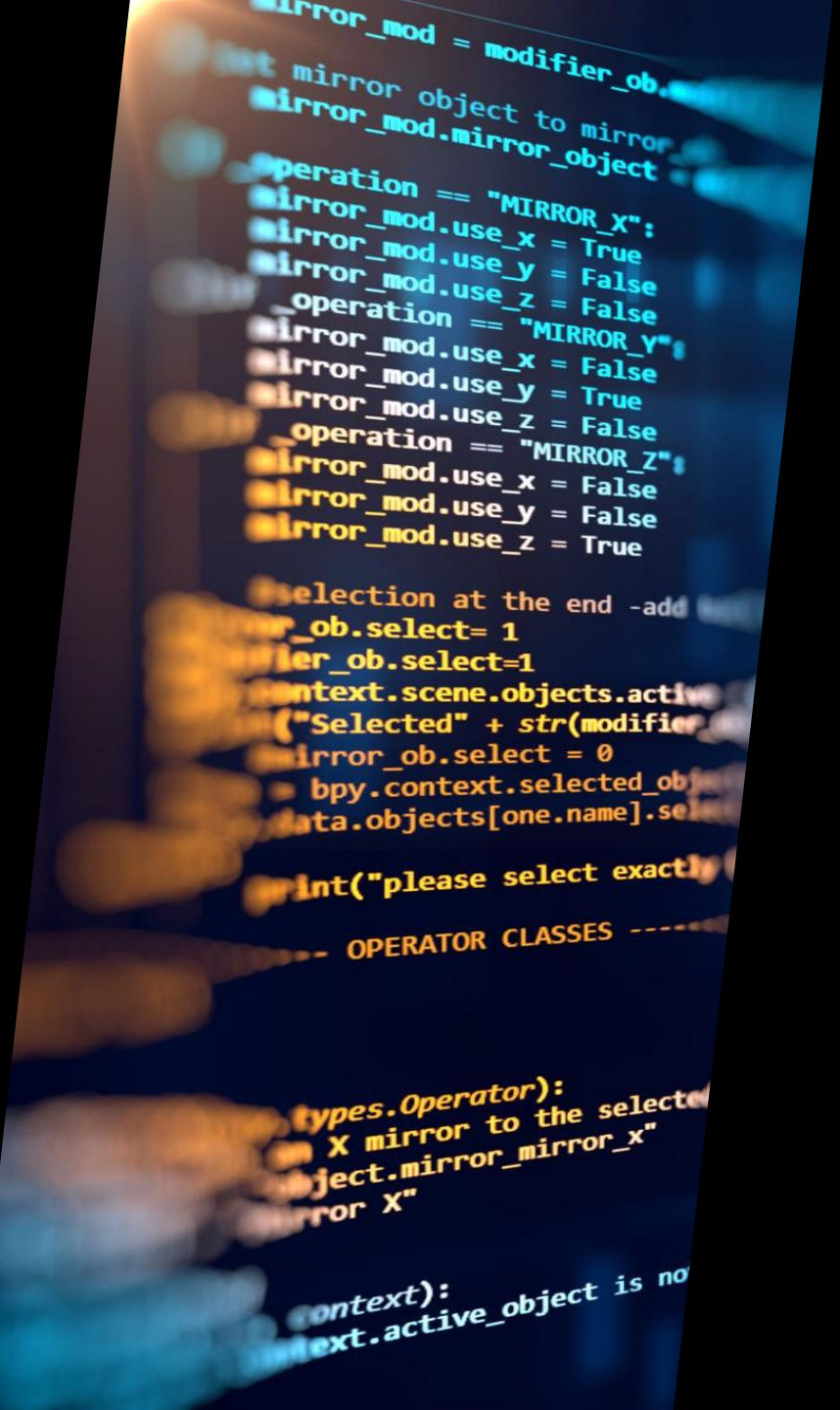
- **System:** Handles the backend processes of file encryption and decryption.

Screen Shots




File Encryptor/Decryptor
File Encryptor/Decryptor
Encrypt File
Decrypt File
Success
File encrypted successfully!
Key: sorAjW2Ro2CgjGw5QqlXLzJvZ52wUbbpDIVCiqkevE4=
OK
PS C:\Users\Admin\OneDrive\Desktop\portfolio> python ss.py
C:\Users\Admin\AppData\Local\Programs\Python\Python312\python.exe: can't open file 'C:\\Users\\Admin\\OneDrive\\Desktop\\portfolio\\ss.py': [Errno 2] No such file or directory


Cipher Hub
Cipher Hub
File Encryptor/Decryptor
Encrypt File
Success
File encrypted successfully!
Key: vf3viqYVS-C2yyLIK5e2t6-3N-7CZ6yhvPsGhKCxYms=
OK



▶**Home Screen Description** : The starting interface for the application . Contains buttons for "Encrypt File" and "Decrypt File.

▶*Flow :* User selects the desired operation to proceed.

▶ **File Selection Screen Description :** Allows users to browse and select files from their system . Includes a confirmation option after file selection.

▶ **Key Generation Screen Description :** Generates a unique encryption key . Displays the generated key and provides options to save/download it .

▶**Flow:** Key is generated automatically when encrypting files.

Conclusion and Future scope

Conclusion: The file encryptor and Decryptor project successfully demonstrates the ability to secure and restore files using encryption and decryption algorithms. It ensures that sensitive data remains protected from unauthorized access and can only be decrypted with the correct key.

Future Scope: Support for More Algorithms Implement additional encryption methods like AES or RSA

Cloud Integration: Extend the project to encrypt files stored on cloud platforms.

User Interface Improvements: Enhance the interface for easier use, especially for non-technical users.

Optimization: Improve the performance of encryption/decryption for larger files.

References

- **Stallings, W. (2017). Cryptography and Network Security:** Principles and Practice (7th ed.). Pearson.This textbook provides an in-depth understanding of cryptographic algorithms and network security, useful for understanding the foundations of encryption and decryption.
- **Barker, E., & Barker, W. (2015). NIST Special Publication 800-57**: Recommendation for Key Management (Revision 4). National Institute of Standards and Technology.A standard reference for key management in encryption, which is essential for securely managing cryptographic keys.
- **Chander, A., & Sethi, A. (2020). Understanding Cryptography:** A Textbook for Students and Practitioners. Springer.A practical guide to cryptography, offering insights into various encryption techniques and their applications.

QUERIES

1. How does AES encryption work?

→ This query can help explore the Advanced Encryption Standard (AES) algorithm, its working principle, and how it can be implemented in a file encryption system.

2. What is the difference between symmetric and asymmetric encryption?

→ Understanding this distinction is crucial for choosing the right algorithm for file encryption and decryption.

3. How can I securely manage encryption keys?

→ This query can help explore key management strategies, such as using key stores or key derivation functions, to ensure that encryption keys are kept secure.

THANK YOU